



1

Feature Selection

Incorporating useless features can be counter productive
Often achieve better performance by removing features
Can be difficult to know a priori which features to use
Exists weighting schemes for features based on the correlation of feature with target value
Can test the usefulness of features by measuring generalisation performance

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Feature Generation/ Reduction

For some inputs such as images or audio signals the number of features are too large

In this case the inputs are often reduced, e.g.

- Extract signatures of signals such as energy in different spectrum bands

- Can search for features in images such as lines

One general technique to reducing complexity is to use PCA

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Measuring Generalisation Performance

Recall, we want to predict **unseen** data

Useful to measure performance

- Often need to set parameters, select features, etc.

- Want to know what errors you expect

You **cannot** use data that you have trained on!

WHY???

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Estimating generalisation performance is important for choosing parameters (degree of regularisation, number of learning steps, complexity of model)

However, you should know how reliable the generalisation performance is—otherwise you are tuning parameters to noise

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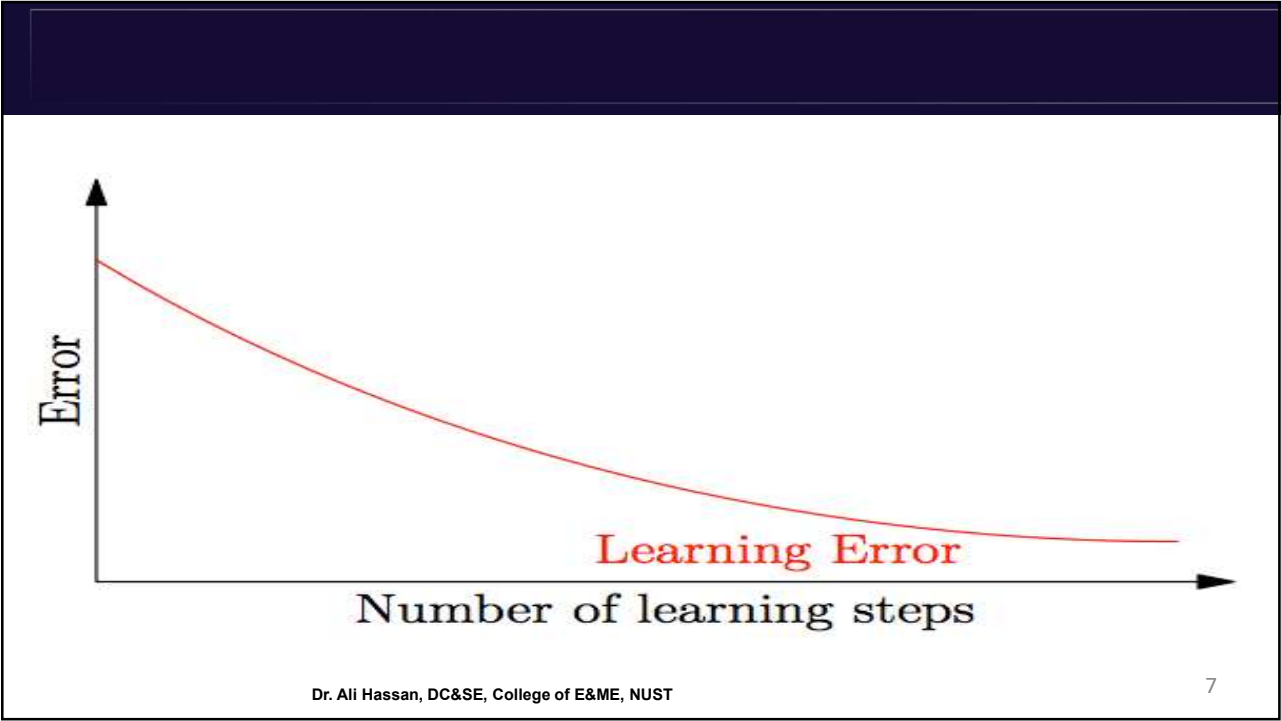
Over Fitting

As the number of learning steps increase the learning error falls but after some time the generalisation error tends to rise due to over-fitting

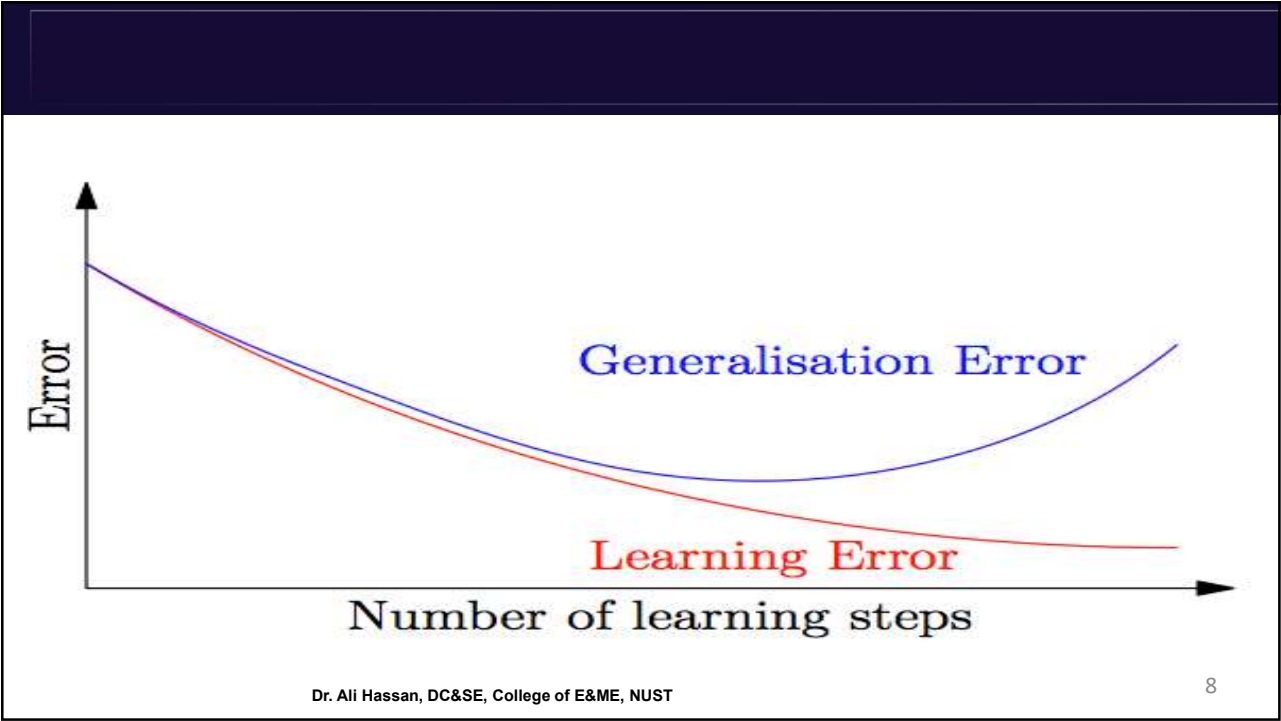
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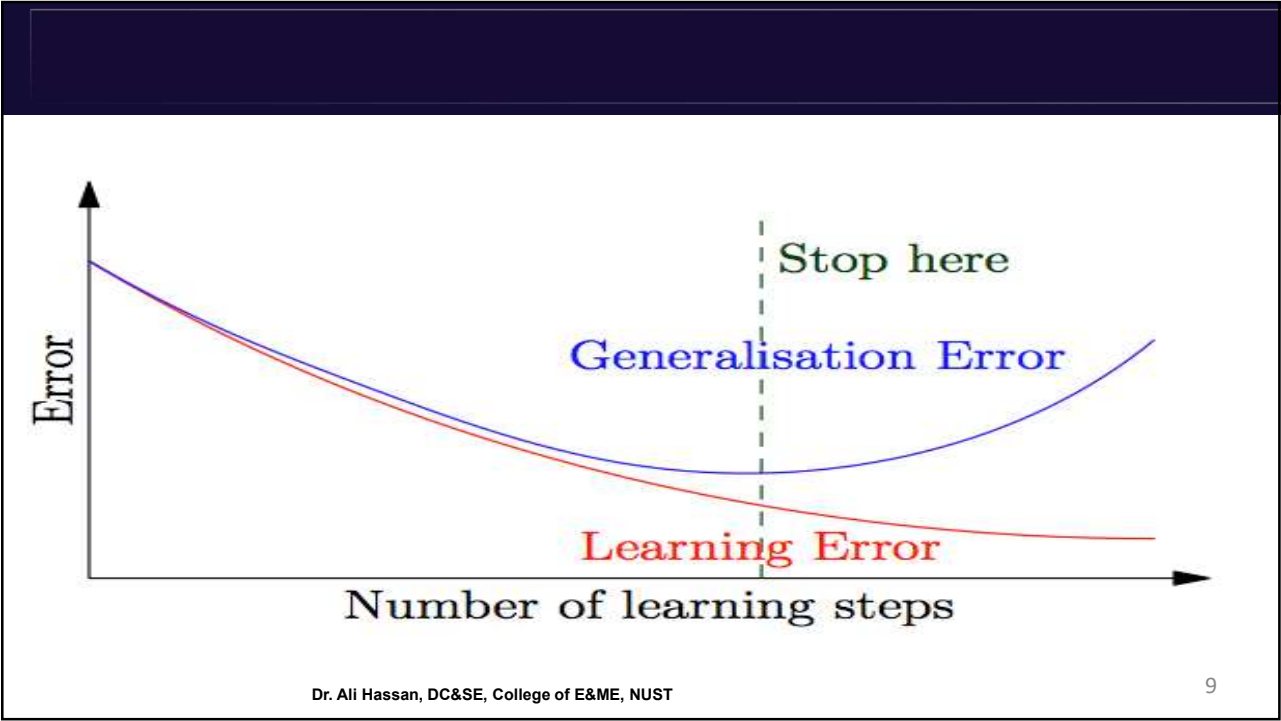
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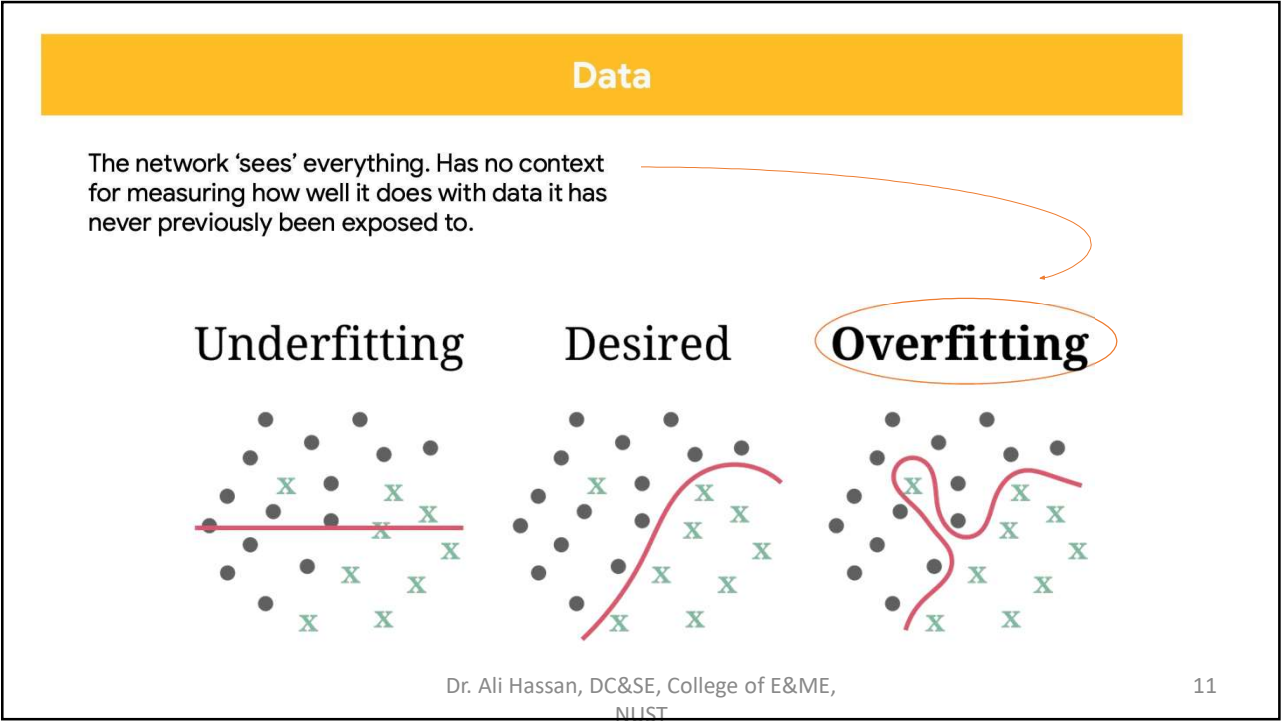
9

Data

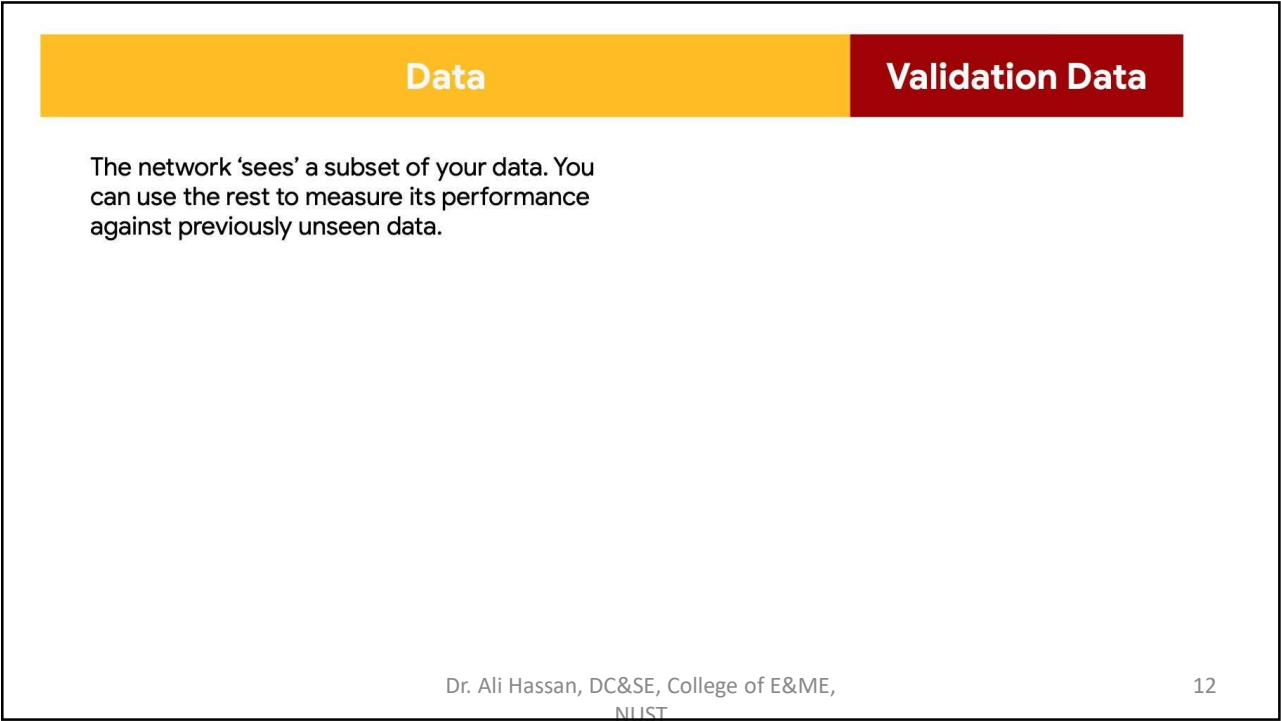
The network 'sees' everything. Has no context for measuring how well it does with data it has never previously been exposed to.

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Data

Validation Data

Test Data

The network 'sees' a subset of your data. You can use an unseen subset to measure its accuracy while training (validation), and then another subset to measure its accuracy after it's finished training (test).

↓

Is used to evaluate the current training epoch

↓

Is used to evaluate the final model after training

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Data

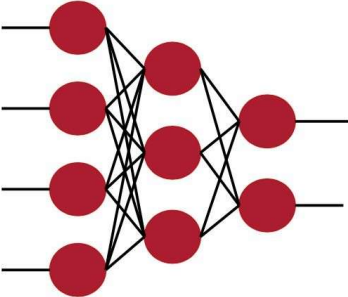
Validation Data

Test Data

Accuracy: 0.999

Accuracy: 0.920

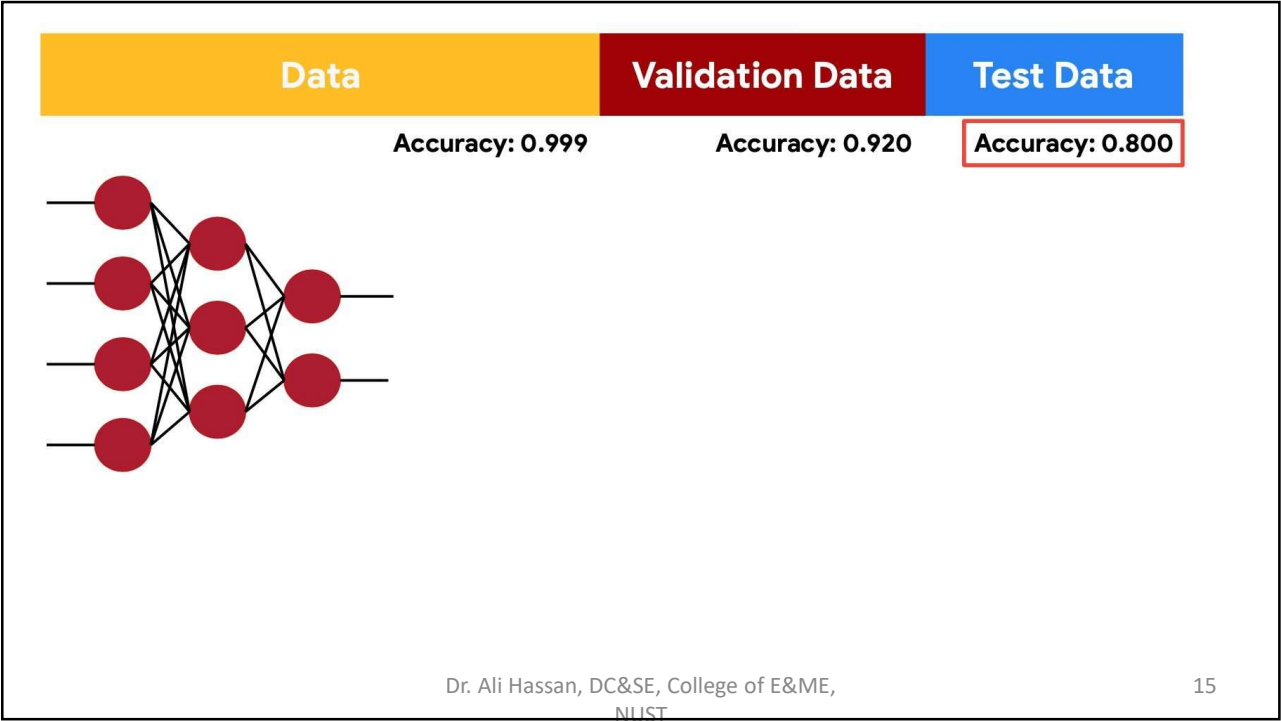
Accuracy: 0.800



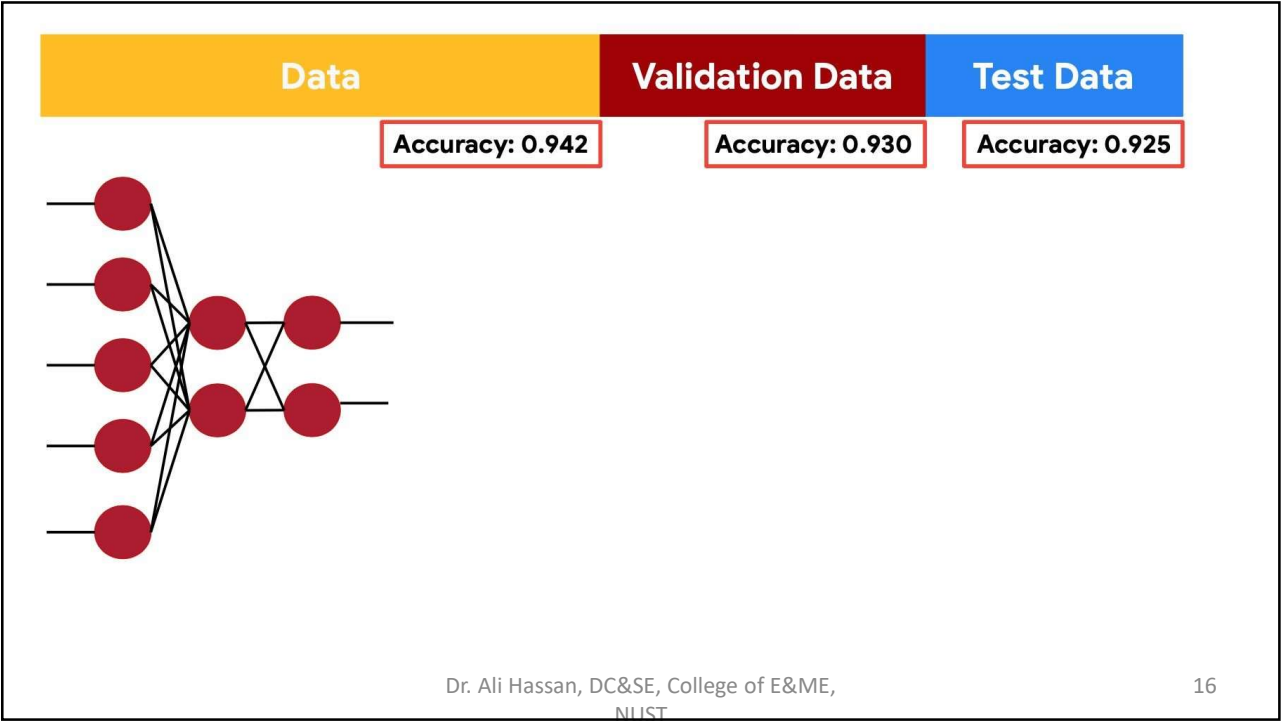
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```
1 data = tf.keras.datasets.mnist
2
3 (tt_images, tt_labels), (test_images, test_labels) = data.load_data()

1 print(tt_images.shape)
2 print(tt_labels.shape)

(60000, 28, 28)
(60000,)

1 print(test_images.shape)
2 print(test_labels.shape)

(10000, 28, 28)
(10000,)
```

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```
1 val_images = tt_images[:10000]
2 val_labels = tt_labels[:10000]

1 train_images = tt_images[10000:]
2 train_labels = tt_labels[10000:]

1 print(train_images.shape)
2 print(train_labels.shape)

(50000, 28, 28)
(50000,)

1 print(val_images.shape)
2 print(val_labels.shape)

(10000, 28, 28)
(10000,)
```

Split tt data in:

- train (50,000) and,
- validation (10,000)

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```

1 history = model.fit(
2     train_images,
3     train_labels,
4     epochs=20,
5     validation_data=(val_images, val_labels)
6 )

```

You could leave the training data with all samples, and alternatively use:

- validation_split=0.1 instead of `validation_data=(val_images, val_labels)`.

In this case, TF will split the validation data by itself.

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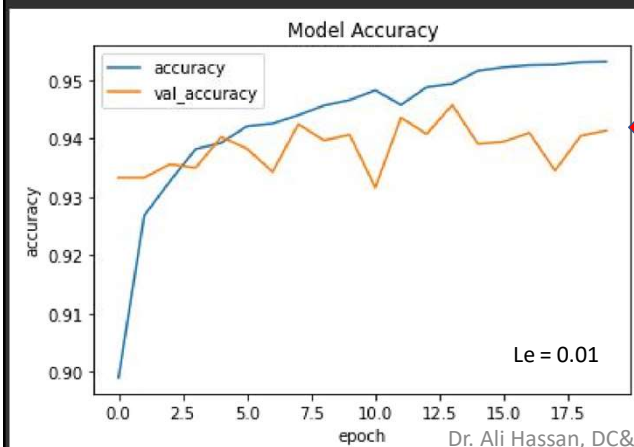
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```

plt.plot(history.history['accuracy'], label='accuracy')
plt.plot(history.history['val_accuracy'], label='val_accuracy')
plt.title('Model Accuracy')
plt.ylabel('accuracy')
plt.xlabel('epoch')
plt.legend(loc='upper left')
plt.show()

```



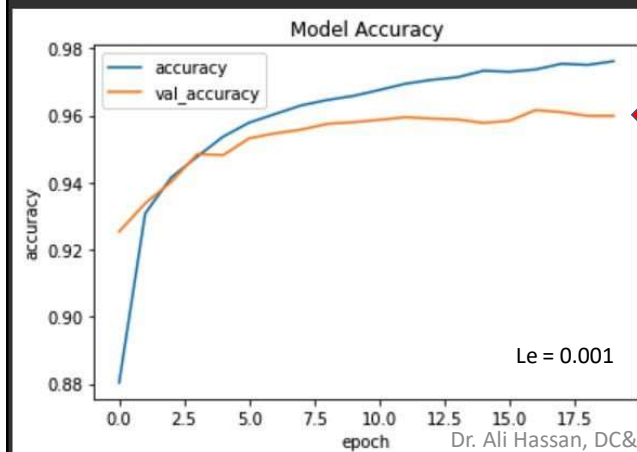
If validation accuracy seems “unstable”, could be that Learning Rate is high (try to reduce it).

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```
plt.plot(history.history['accuracy'], label='accuracy')
plt.plot(history.history['val_accuracy'], label='val_accuracy')
plt.title('Model Accuracy')
plt.ylabel('accuracy')
plt.xlabel('epoch')
plt.legend(loc='upper left')
plt.show()
```



If validation accuracy goes down (or became stable) , even if train accuracy goes up, means that probably the model is overfitting. In this case the training (epochs) should terminate.

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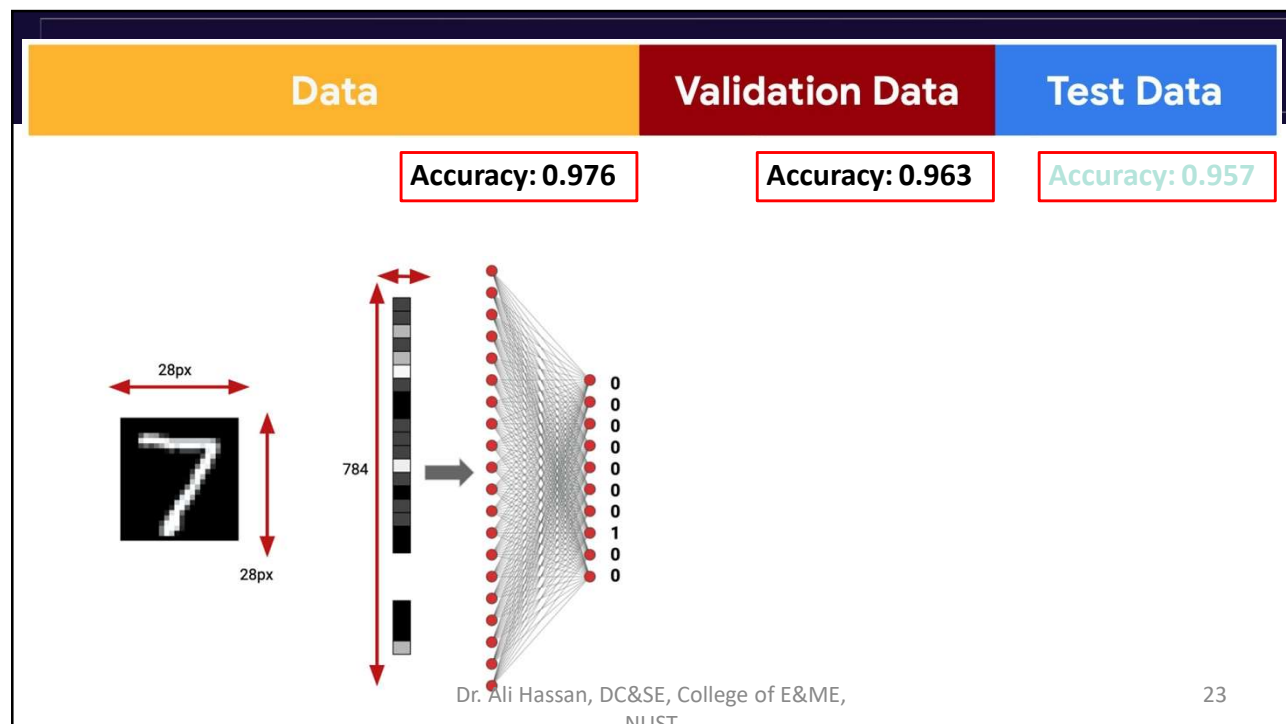
```
model.evaluate(test_images, test_labels)
```

```
313/313 [=====] - 1s 2ms/step - loss: 0.1495 - accuracy: 0.9569
```

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In summary

- Training Data** -> Used to train **model parameters**
- Validation Data** -> Used to determine what **model hyperparameters** to adjust (and re-training)
- Test Data** -> Used to get **model final performance metric**

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Going Further

Classification Model Performance Metrics

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Class = [1]



Class = [0]

actual = [1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0]

prediction = [0, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1]

Data
(Actual)

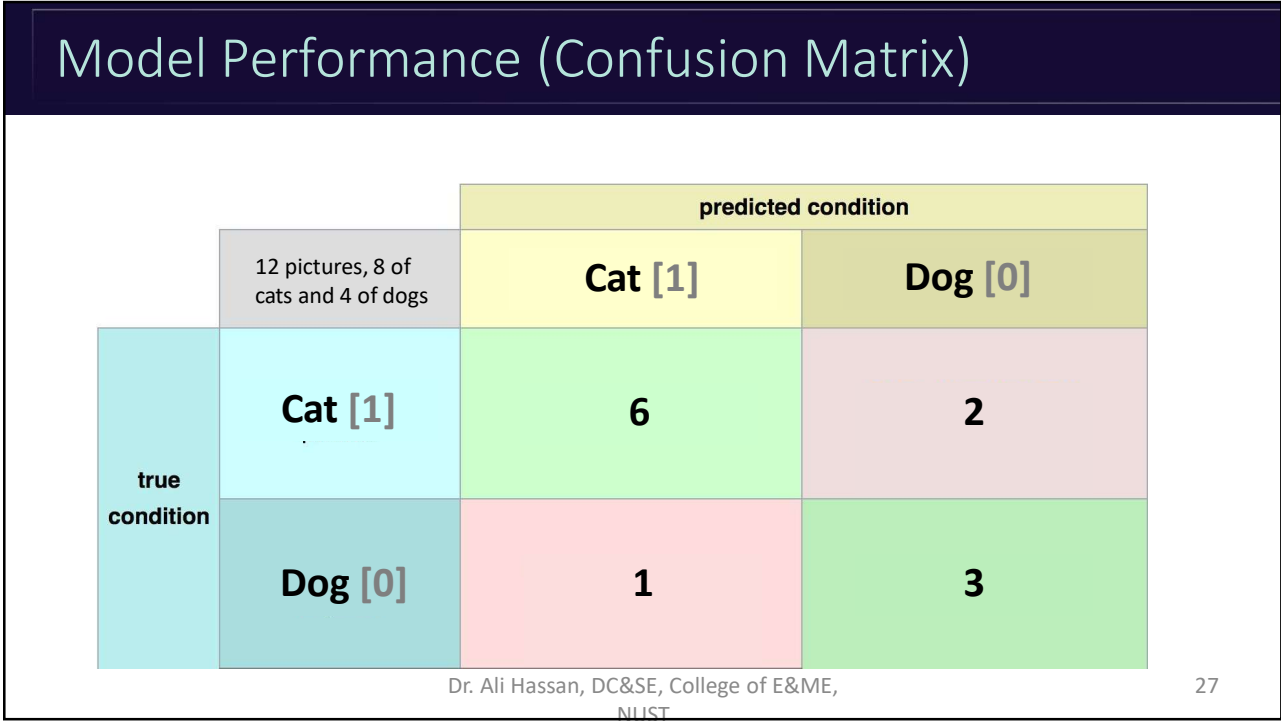
Model

Inferences
(Prediction)

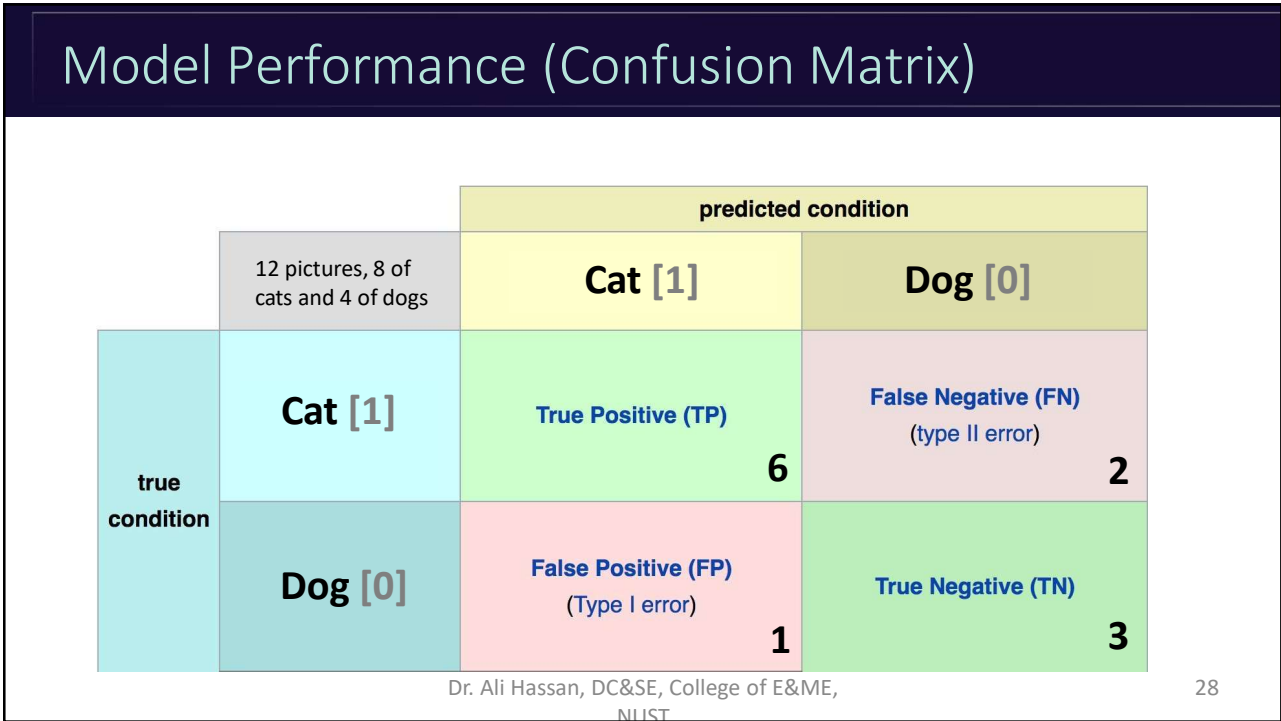
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Model Performance (Confusion Matrix)

		predicted condition	
		prediction positive (PP)	prediction negative (PN)
true condition	condition positive (P)	True Positive (TP)	False Negative (FN) (type II error)
	condition negative (N)	False Positive (FP) (Type I error)	True Negative (TN)

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https://en.wikipedia.org/wiki/Confusion_matrix

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Type I error (false positive)



A doctor in a white coat is speaking to an older man. A speech bubble from the doctor says "You're pregnant". The man is looking down with a thoughtful or perhaps surprised expression.

Type II error (false negative)



A doctor in a white coat is examining a pregnant woman lying in a hospital bed. A speech bubble from the doctor says "You're not pregnant". The woman is looking up at the doctor with a concerned expression.

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Precision

VS.

Accuracy



High Precision, High Accuracy



Low Precision, High Accuracy



High Precision, Low Accuracy



Low Precision, Low Accuracy

In a set of measurements:

- **Accuracy** is closeness of the measurements to a specific value
- **Precision** is the closeness of the measurements to each other.

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Accuracy , Precision and Recall

Accuracy = $\frac{TP + TN}{(P + N)}$ = $\frac{TP + TN}{(TP + TN + FP + FN)}$ = $\frac{6 + 3}{(6 + 3 + 1 + 2)}$ = $\frac{9}{12}$ = 0.75

Precision = $\frac{TP}{(TP + FP)}$ = $\frac{6}{(6 + 1)}$ = $\frac{6}{7}$ = 0.86

$\frac{\text{Total Positive}}{\text{Total Predict Positive}}$

Recall = $\frac{TP}{(TP + FN)}$ = $\frac{6}{(6 + 2)}$ = $\frac{6}{8}$ = 0.75

(or Sensitivity)

$\frac{\text{Total Positive}}{\text{Total Actual Positive}}$

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F1-Score

F1 = 2 x $\frac{\text{Precision} * \text{Recall}}{(\text{Precision} + \text{Recall})}$

F1 = 2 x $\frac{(0.86 * 0.75)}{(0.86 + 0.75)} = 2 \times \frac{0.65}{1.61} = 0.80$

The F1-score is a way of combining the **precision** and **recall** of the model

relevant elements

false negatives

true negatives

true positives

false positives

selected elements

How many selected items are relevant?

Precision = $\frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$

How many relevant items are selected?

Recall = $\frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$

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<https://en.wikipedia.org/wiki/F-score#Formulation>

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Pts without the disease

Pts with disease

Test Result

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Choosing the threshold

Where should we set the **threshold**

We could choose the **equal error rate** point where the errors in positive training set equals the errors in the negative training set

Data classes may be very imbalanced (e.g. $|D^+| \ll |D^-|$)

In many applications we might be risk averse to false positives or false negatives

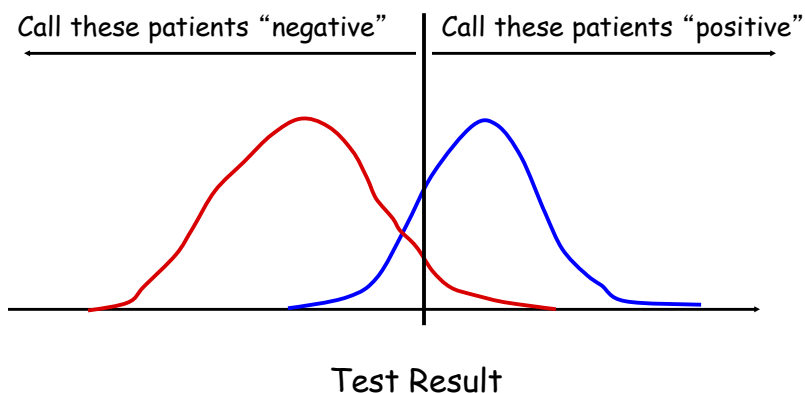
Want to see all the options

The **receiver operating characteristic (ROC)** curve is a standard way to test this

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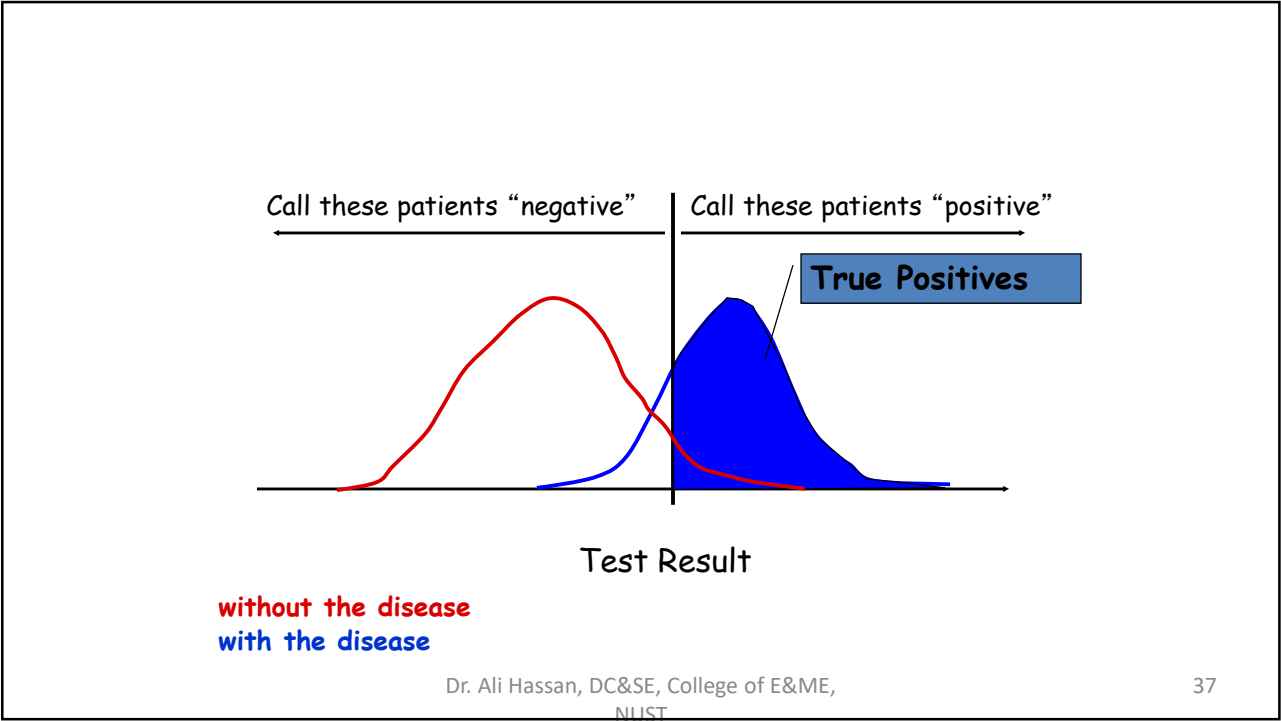
35



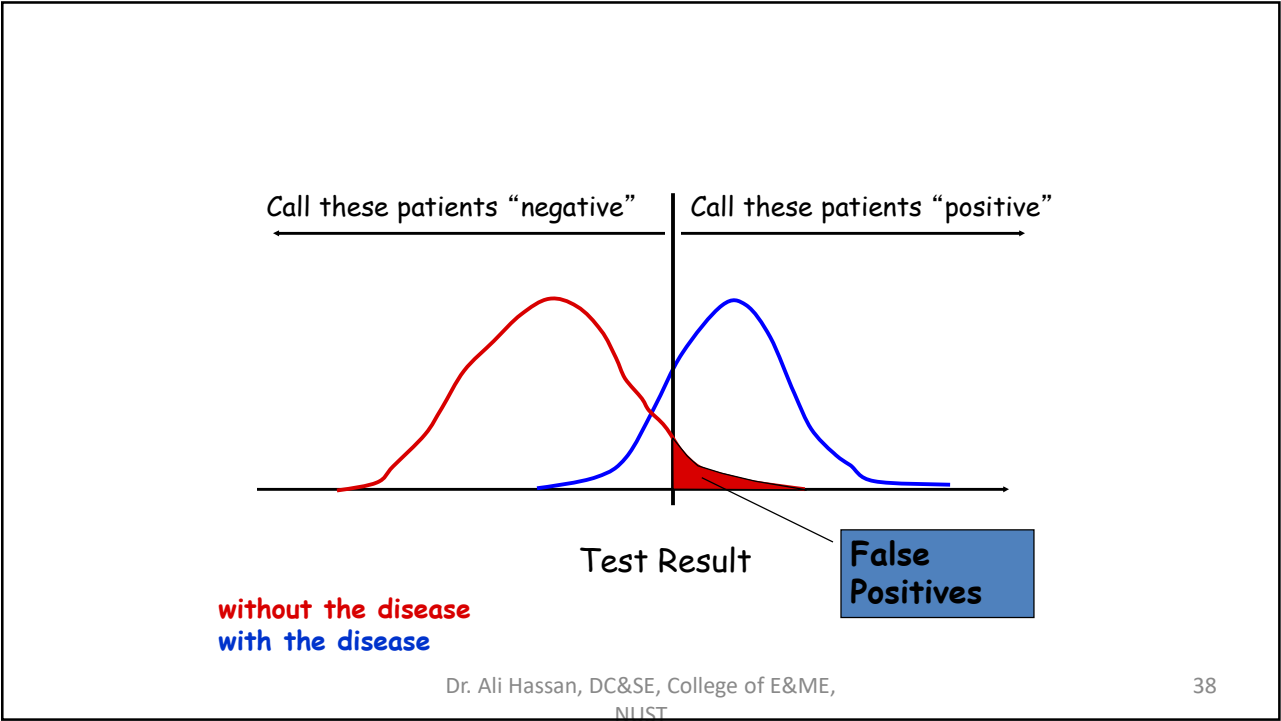
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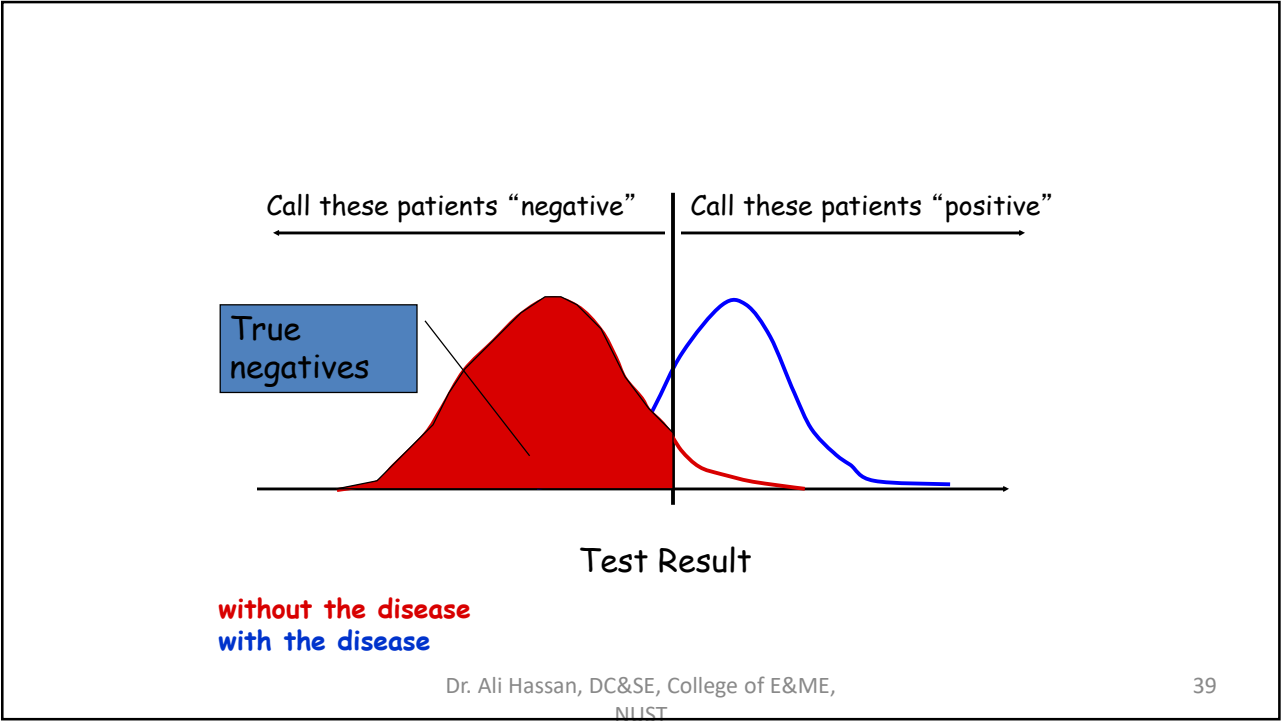
36



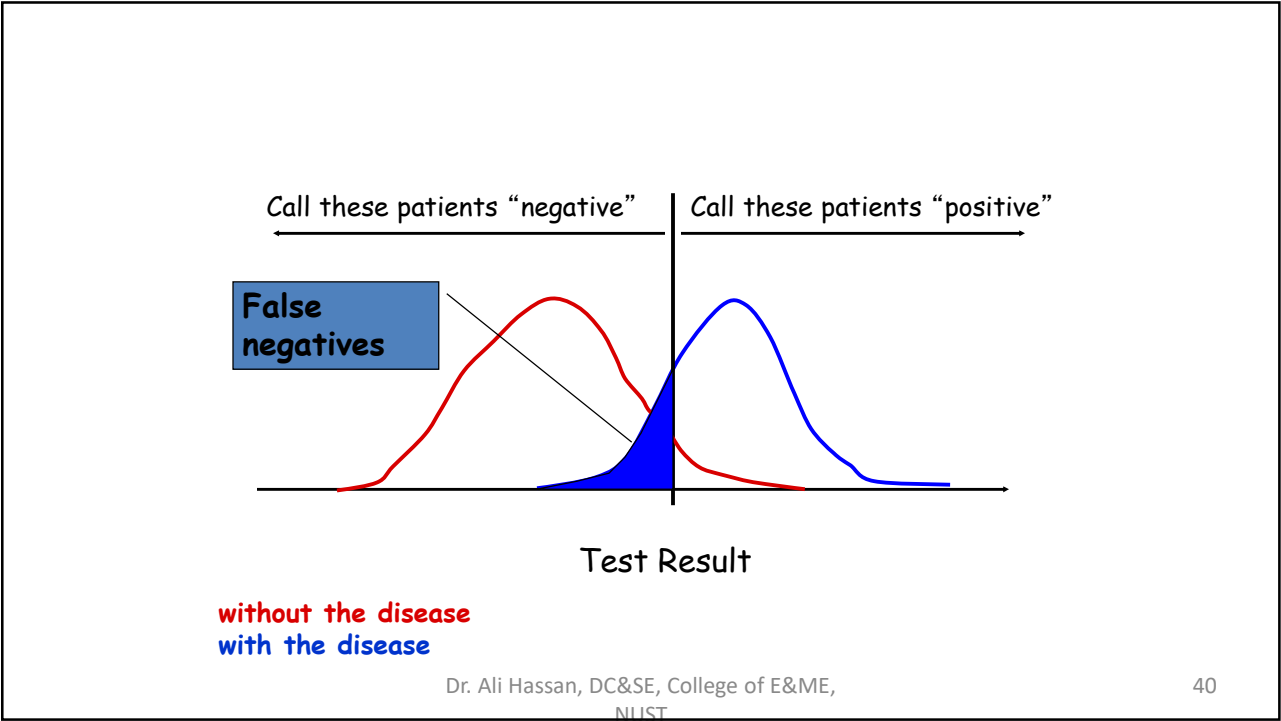
37



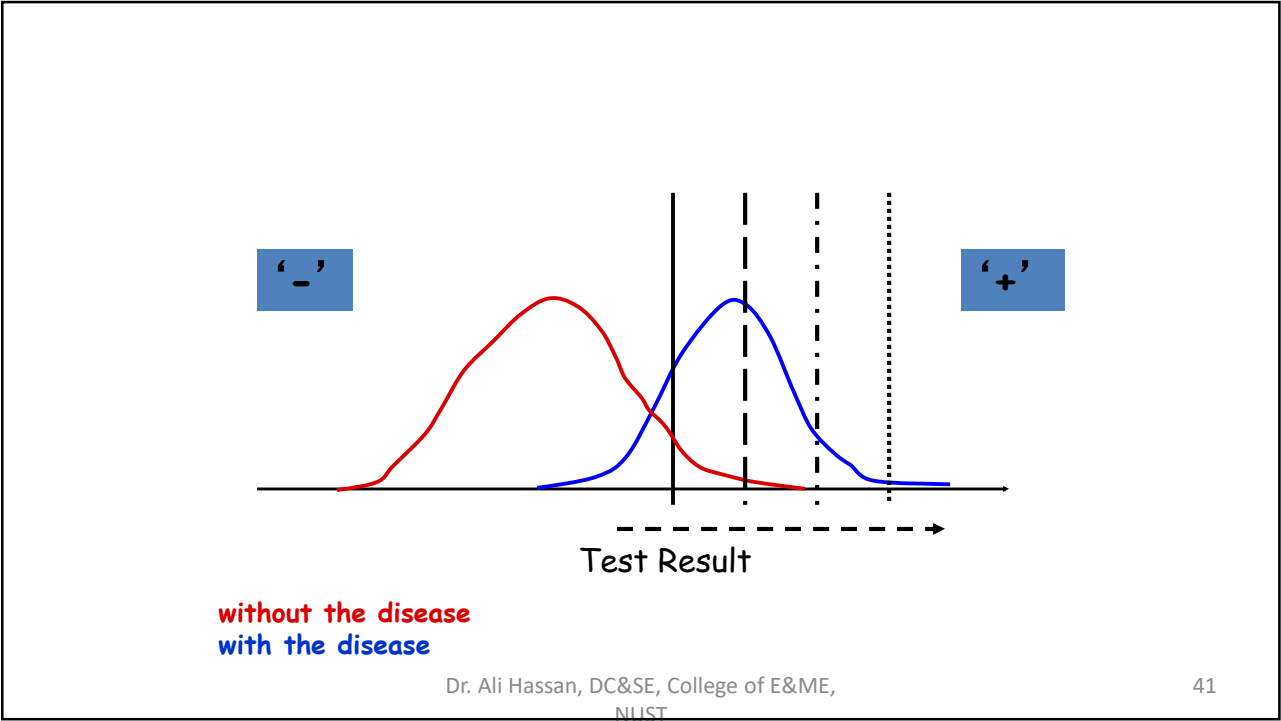
38



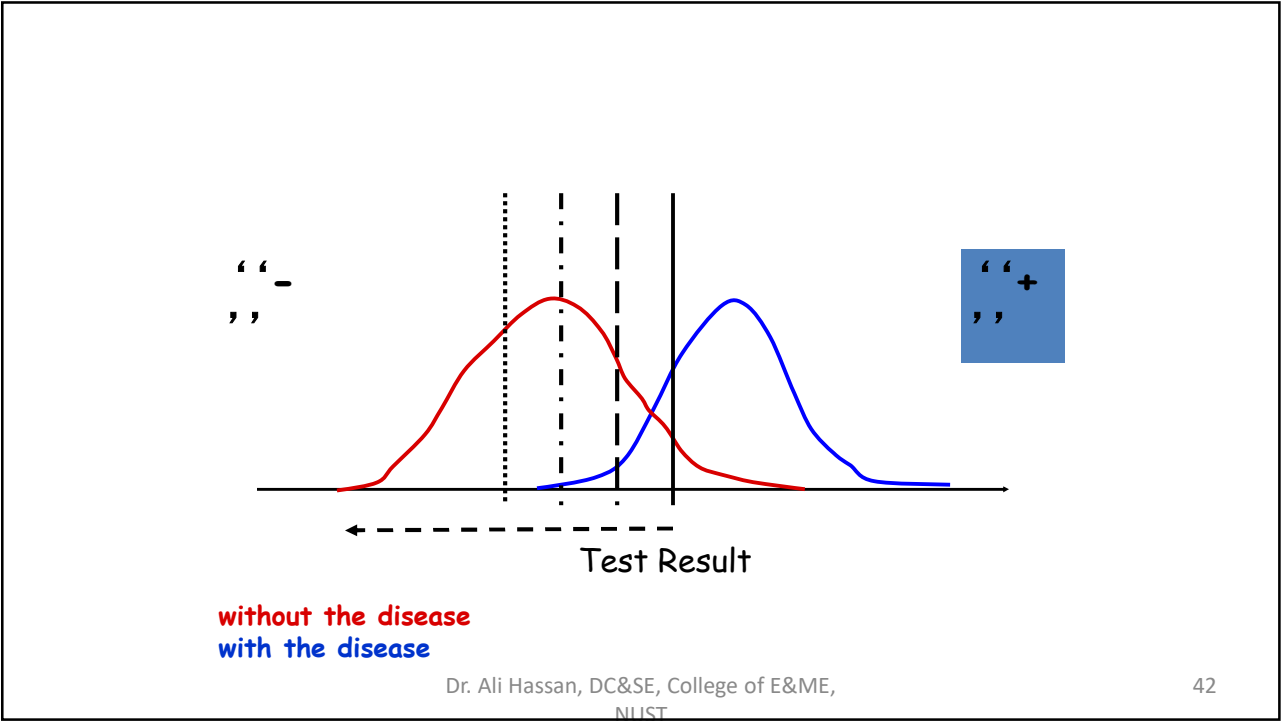
39



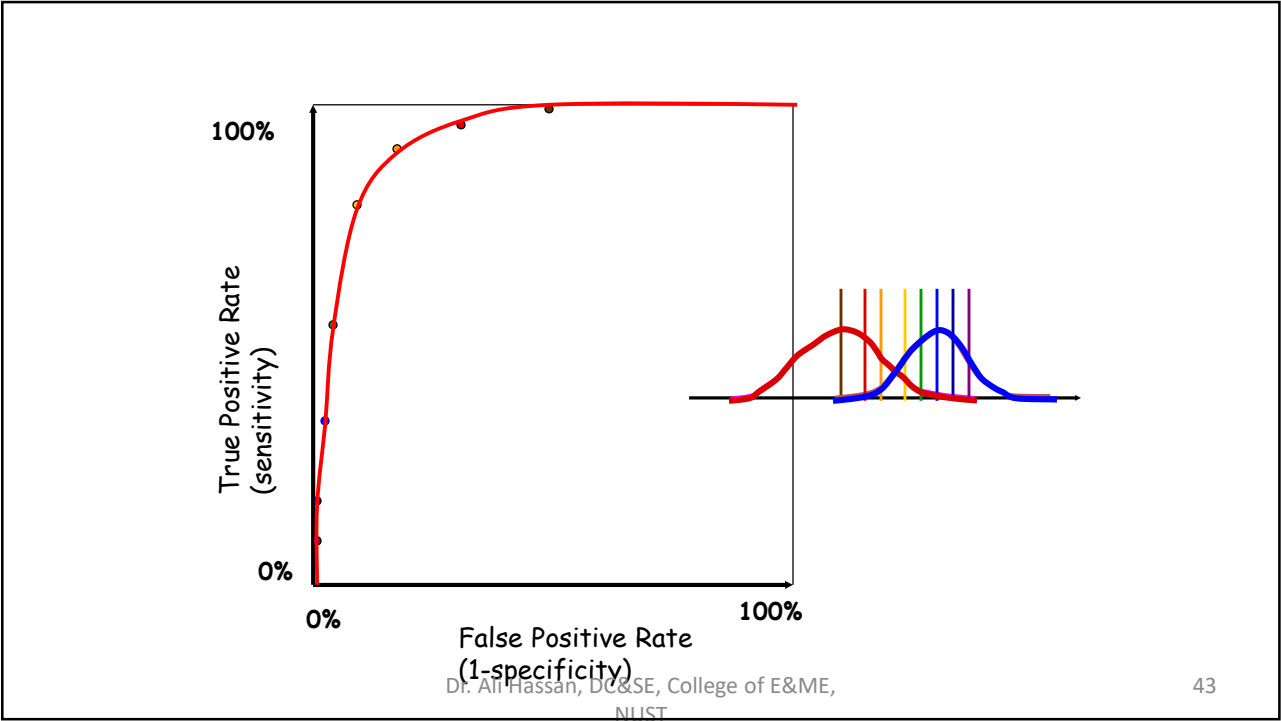
40



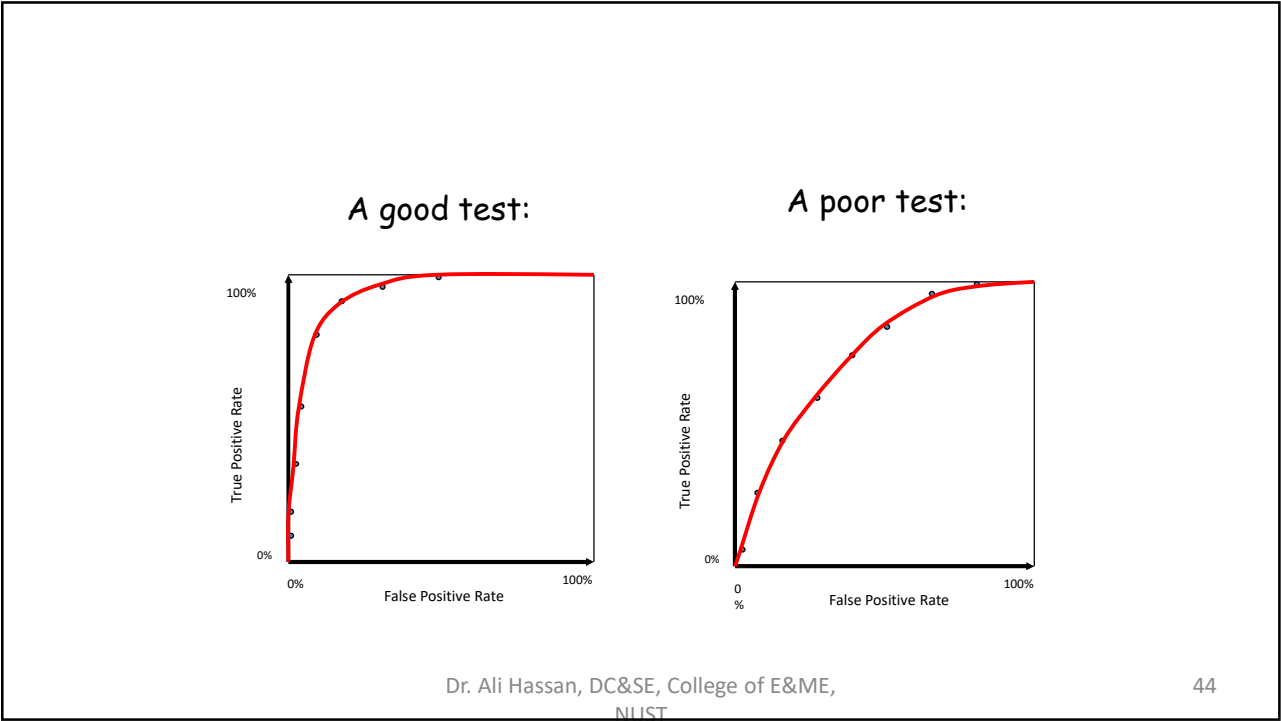
41



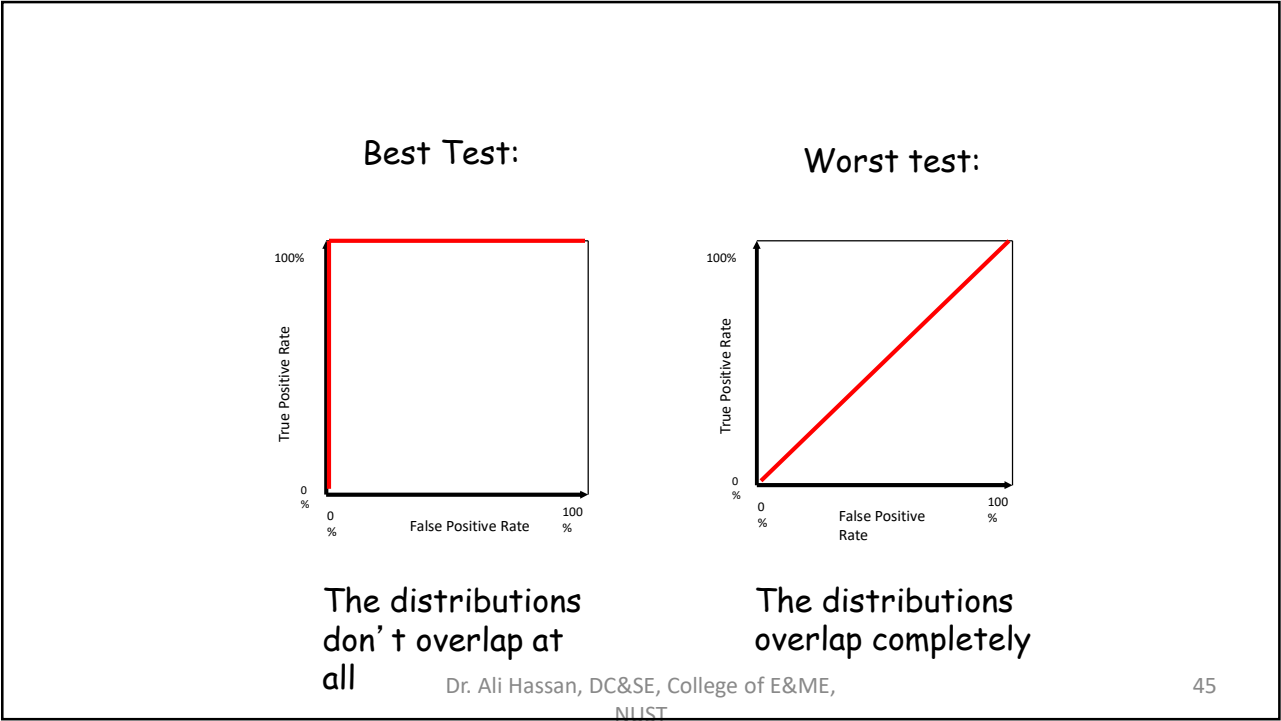
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Area under ROC curve (AUC)

Overall measure of test performance

Comparisons between two tests based on differences between (estimated) AUC

ROC curves can be calculated for most (but not all) binary classifiers

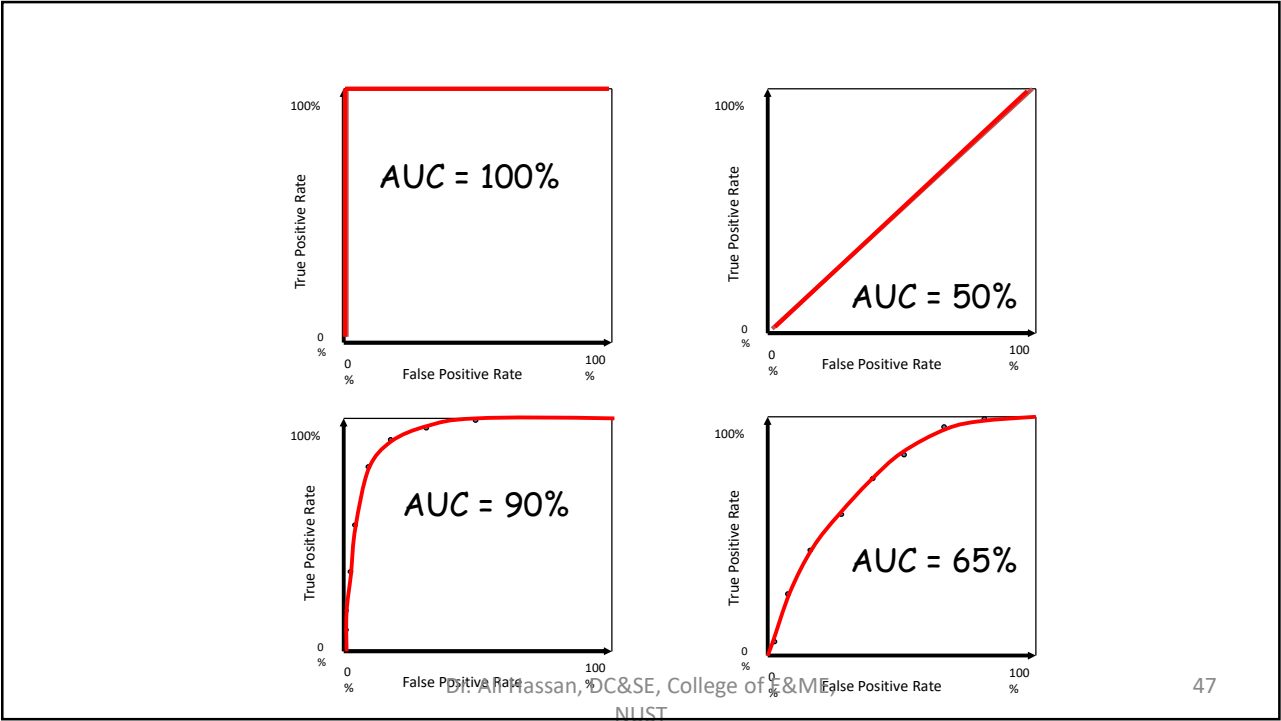
The area under the ROC curve is now the standard method for measuring performance in binary classification problems

This should be between 0.5 (random) to 1 (perfect)

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Interpretation of AUC

AUC can be interpreted as the probability that the test result from a randomly chosen diseased individual is more indicative of disease than that from a randomly chosen non-diseased individual: $P(X_i \geq X_j \mid D_i = 1, D_j = 0)$

So can think of this as a nonparametric distance between disease/nondisease test results

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Problems with AUC

No clinically relevant meaning

A lot of the area is coming from the range of *large false positive* values, no one cares what's going on in that region (need to examine restricted regions)

The curves might *cross*, so that there might be a meaningful difference in performance that is not picked up by AUC

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Examples using ROC analysis

Threshold selection for 'tuning' an already trained classifier (e.g. neural nets)

Defining signal thresholds in DNA microarrays (Bilban *et al.*)

Comparing test statistics for identifying differentially expressed genes in replicated microarray data (Lönnstedt and Speed)

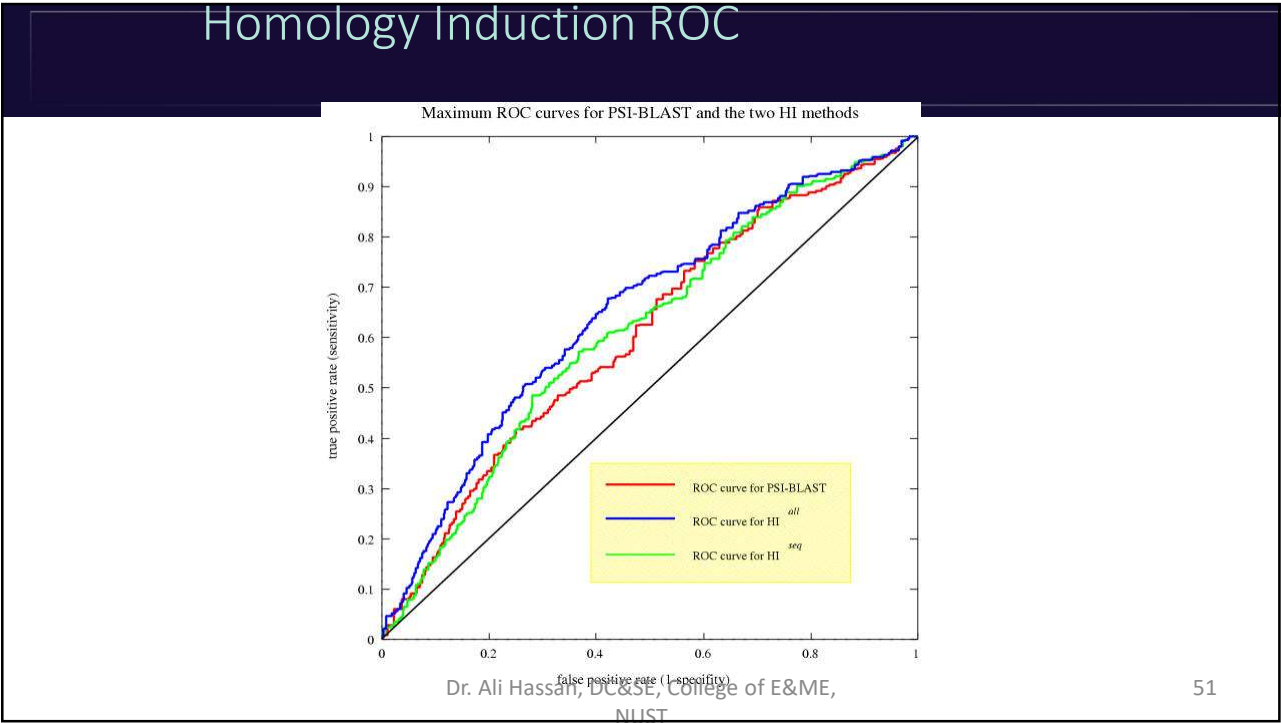
Assessing performance of different protein prediction algorithms (Tang *et al.*)

Inferring protein homology (Karwath and King)

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Classification Report

Code Time!



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```
1 print(classification_report(actual, prediction, target_names=target_names))
```

	precision	recall	f1 score	support
Dogs	0.60	0.75	0.67	4
Cats	0.86	0.75	0.80	8
accuracy			0.75	12
macro avg	0.73	0.75	0.73	12
weighted avg	0.77	0.75	0.76	12

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Cross Validation

We can solve our dilemma by repeating the training many times on different partitioning
This is known as K-fold cross validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

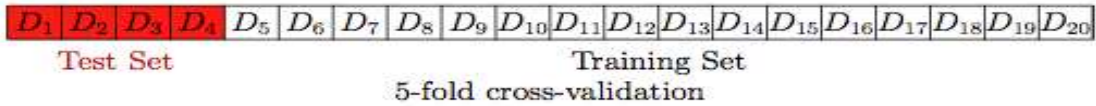
$$D = \{D_i\}_{i=1}^P \quad D_i = (x_i, y_i)$$

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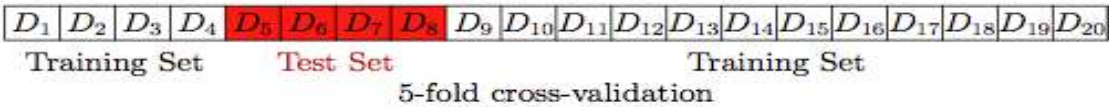
Cross Validation



$E_g =$ 5.1

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Cross Validation



$E_g =$ 3.7

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
Training Set								Test Set				Training Set							

5-fold cross-validation

$E_g =$ 4.6

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
Training Set												Test Set				Training Set			

5-fold cross-validation

$E_g =$ 4.6

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test Set

Training Set

10-fold cross-validation

$E_g = 5.8$

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Test Set

Training Set

10-fold cross-validation

$E_g = 1.8$

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
Training Set				Test Set		Training Set													

10-fold cross-validation

$E_g =$ 4.8

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
Training Set						Test Set		Training Set											

10-fold cross-validation

$E_g =$ 3.6

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
Training Set								Test Set		Training Set									

10-fold cross-validation

$E_g = 7.4$

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
Training Set										Test Set		Training Set							

10-fold cross-validation

$E_g = 0.99$

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
Training Set												Test Set		Training Set					

10-fold cross-validation

$E_g =$ 4.5

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
Training Set														Test Set		Training Set			

10-fold cross-validation

$E_g =$ 5.4

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
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Training Set

10-fold cross-validation

Test Set

$E_g =$

6.2

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Training Set

10-fold cross-validation

Test Set

$E_g =$

2.7

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

$$\langle E_g \rangle = \frac{5.8 + 1.8 + 4.8 + 3.6 + 7.4 + 0.99 + 4.5 + 5.4 + 6.2 + 2.7}{10} = 4.3$$

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test

Leave-one-out cross-validation

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test

Leave-one-out cross-validation

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test

Leave-one-out cross-validation

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test

Leave-one-out cross-validation

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test

Leave-one-out cross-validation

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Cross Validation

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

$$\langle E_g \rangle = 3.9$$

- Leave-one-out cross-validation is extreme case

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Price of Cross Validation

Cross-validation is computationally expensive (K-fold cross-validation requires K times as much work)

There are attempts at estimating generalisation error more cheaply (bootstrapping) methods, but these are not very accurate

Cross-validation is only necessary when you have little data

Re-running code is usually cheap compared with writing the code

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Reading Material

Lecture note by Dr Ali Hassan

Neural Network Tutorial

You Tube Videos

<https://youtu.be/aircAruvnKk>

<https://youtu.be/uXt8qF2Zzfo?si=IPA9PR0tmPTBV4T8>

[Andrew Ng Lectures](#)

<https://youtu.be/1ZhtwlnuOD0>

**Neural Networks and Deep Learning by Michael
Nielson**

<http://neuralnetworksanddeeplearning.com/>

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